

PRACHI MARUTI PATIL

Full Stack Developer · AI/ML Engineer · Computer Engineering, Data Science Honours
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PROFESSIONAL SUMMARY

Final-year Computer Engineering student specialising in Data Science (CGPA 9.79/10) with proven experience building AI-led, production-ready systems across full-stack engineering, machine learning, and cloud data platforms. Contributed to intelligent automation research at BARC (Govt. of India) and shipped enterprise-grade software across 3 industry internships. SCOPUS-indexed IEEE researcher and 2× national hackathon finalist. Targeting Prime/Digital engineering track roles where deep technical delivery meets AI-enabled transformation at scale.

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning · Computer Vision · YOLOv8 · TensorFlow · Scikit-learn · Generative AI · SHAP · Pandas

Full-Stack Engineering: React.js · Next.js · Node.js · Express.js · REST APIs · WebSocket · JWT Authentication · Three.js · HTML5 · CSS3

Data Engineering: Data Pipelines · Web Scraping · Data Cleaning · SQL · MongoDB · MySQL · Firebase · Data Visualisation

Cloud & DevOps: Azure Cloud Platform · Vercel CDN · Git / GitHub · CI/CD Fundamentals · MLOps Basics

Languages: Python · JavaScript · Java · SQL · HTML5 · CSS3

EXPERIENCE

Bhabha Atomic Research Centre (BARC) — Project Trainee

Nov 2025 — Jan 2026

Security Electronics & Cyber Technology Division, Trombay · Govt. of India

- Engineered an intelligent automation system achieving 90%+ defect detection accuracy at 10 images/sec, classifying 4 weld defect types (pores, clots, thin cracks, large cracks) using Python and Emgu CV at a national nuclear research facility and GUI for interaction.

Vidyarthimitra — Full Stack Development Intern

May 2025 — Jun 2025

Mumbai, India

- Engineered a machine learning-powered college recommendation platform (React + MySQL) using data engineering pipelines to scrape and process data from 500+ sources, delivering personalised recommendations across 5,000+ college-course combinations.

Paarvedan Foundation — Junior Developer

Jan 2025 — Mar 2025

Mumbai, India

- Expanded an ERP system serving 500+ active users by shipping 8 new features and resolving 25+ critical bugs; rebuilt the community website frontend, measurably reducing page load time through asset optimisation and UI restructuring.

PROJECTS

VigilAI: Smart Surveillance Event Detection & Analysis System [GitHub ↗](#)

React.js · Node.js · Python · YOLOv8

- Developed an enterprise-grade AI-led surveillance platform using deep learning (YOLOv8, ~80% mAP) at 24–27 fps, detecting 4+ event types via rule-based intelligent automation; 5-min video processed in under 3 min with a multi-service analytics dashboard.

Inventory Management System [Live ↗](#)

React.js · Node.js · MongoDB · Azure

- Architected a full-stack inventory platform on Azure cloud, designing RESTful Node.js APIs and implementing real-time tracking to eliminate manual stock reconciliation; monitored live API performance via Azure Metrics dashboard.

FlipFrame — 3D Orthographic Viewer [Live ↗](#)

Next.js · Three.js · Vercel

- Delivered an interactive 3D visualisation tool for engineering students achieving 60 fps performance and sub-2s load time via SSR, with bundle size optimised under 500 KB and deployed globally on Vercel CDN.

Vocabify — Vocabulary Learning App [GitHub ↗](#)

React.js · Node.js · MongoDB · JWT

- Built a multilingual vocabulary platform supporting 10+ languages with a spaced repetition algorithm, cutting MongoDB query latency by 73% (450 ms → 120 ms) through targeted index optimisation.

RESEARCH PUBLICATION

Salary-Sense: A Model Pool Driven Developer Salary Prediction & Career Intelligence System · I5CPS-2026, NIT Puducherry (IEEE Madras Section, Feb 2026) · [SCOPUS Indexed](#)

EDUCATION

Vidyalankar Institute of Technology, Mumbai

2023 — Expected May 2027

B.Tech in Computer Engineering | Honours: Data Science | CGPA: **9.79 / 10.0** | HSC: 93.6% | SSC: 95.6%

CERTIFICATIONS & ACHIEVEMENTS

- National Hackathon Finalist (×2): WCE National Hackathon 2025 & HackOverflow 3.0, Pillai HOC COET — Mar 2025
- International Hackathon Finalist (Team Lead) — TYNET 2.0, Jan 2024 · Top 10 of 500+ teams
- Certifications: Machine Learning with Python — L&T EduTech | Oracle AI Foundations | TCS Data Visualisation (Forage) | Java — NPTEL
- Technical Core, Computer Society of India VIT (2023–24) | Event Core, IEEE VIT Chapter